

Heat wave management plan

This plan is not intended to replace the plan managed by the Ministry of Health regarding health risks, but to complement it.

An interministerial review was conducted following the two heatwave episodes of 2019, as well as the drought episode of 2022. It emerged that, while the health system was not generally put under strain during these two episodes thanks to the activation of the national heatwave plan, these heatwaves impacted the entire population and had major consequences in many areas: closure of schools and nurseries, postponement of school exams, cancellation of sporting and cultural events, disruptions in public transport and great hardship for users, animal mortality, drying up of waterways and difficulties in supplying drinking water, crop fires... It was therefore decided to extend the national heatwave management plan in order to take into account all the impacts of these events.

This plan aims to systematically prepare for the summer period, anticipate the expected arrival of a heatwave, and define the actions to be implemented at the national and local levels to prevent and limit its impacts. It takes into account the specific characteristics of the overseas territories which, due to their geographical and climatic situations, the impact of climate change and rising temperatures, and the lack of infrastructure and its maintenance, face significant heatwaves over extended periods.

The national heatwave management plan is based on the meteorological alert system specific to this phenomenon (currently being adapted for overseas territories) and is organized around actions detailed in the annex targeting four target audiences:

- the general public.
- young people and vulnerable people
- workers and businesses
- Athletes and spectators of cultural events

In addition to this, there is the annual preparation of the players in the electricity sector.

Some of these actions will be implemented annually, prior to periods of high risk of heat waves. Other actions are intended to be implemented when a heat wave is forecast or underway. Episodes of intense heat are generally predicted about a week in advance, thanks to Météo-France's weather forecasting models. This timeframe should be used to alert the various stakeholders involved in risk prevention and management. These actions may be adapted and/or implemented specifically to take into account the particular situation of the overseas territories.

These actions are implemented by the relevant ministries, in conjunction, where appropriate, with the Ministry of the Interior and Overseas Territories, depending on the situation. They complement the national support and crisis management system for public health, led by the Ministry of Health, as described in the interministerial instruction of May 7, 2021, concerning the health management of heat waves in metropolitan France, and in two documents attached thereto. These documents are intended for prefects within the framework of the departmental ORSEC plan, on the one hand, and for mayors within the framework of the municipal emergency plan, on the other.

- Departmental ORSEC Guide S6: "Specific provisions for the health management of heat waves"
Ministry of Health and Solidarity – Directorate General of Health – May 2021

- Guide "Coping with heat waves with your municipal emergency plan – Recommendations and good practices » Ministry of the Interior - Directorate General for Civil Security and Crisis Management - June 2021.

Axis 1. For the general public

Key actions

Action 1. Awareness-raising and communication campaign aimed at the general public

Action 2. Inventory of cool islands and public fountains in municipalities

Action 3. Dissemination of prevention messages in transport

Action 4. Initial assessment of the state-backed start-up "Fresher My City"

Action 5. Sending SMS messages in exceptional circumstances

Action 6. RTE and Enedis strengthen their monitoring and intervention capabilities for periods of heat waves and drought.

Additional actions

Communication aimed at foreign tourists

Axis 2. For young people and vulnerable people

Key actions

Action 7. Mobilizing young people in universal national service to register vulnerable people on municipal registers and in support of climate change awareness actions

Action 8. Raising awareness among guardians/curators of the need to register vulnerable persons on municipal registers

Action 9. Implementation of an annual inspection of rooms and summer comfort equipment in schools and nurseries.

Action 10. Implementation of a control system for rooms and refreshed premises that can accommodate examinations

Axis 3. For workers and businesses

Key actions

Action 11. Creation of a guide for businesses regarding work that can be carried out in offices and communication on appropriate office behavior.

Action 12. Strengthening of labour inspection controls

Action 13. Restriction of the transport of live animals and management of peak activity by rendering companies

Additional actions

Organization by Voies navigables de France (VNF) of a meeting of local user committees

Informing employers of the upcoming heat wave

Regulation of certain activities according to the level of fire risk by the mayor and the prefect

Axis 4. For athletes and spectators of cultural events

Key action

Action 14. Departmental census of sporting and cultural events

Action 15. Invite organizers of sporting and cultural events to raise their awareness of best practices in the event of heat waves.

Additional actions

Information for sports educators

<u>Focus: Preparing stakeholders in the electricity sector</u>

- Availability of the nuclear reactor fleet
- Hydroelectricity
- Annual preparation

Axis 1. For the general public

In order to limit the impact of heat waves on the population, a series of prevention and awareness actions aimed at the general public must be carried out.

To address this issue in housing, the RE2020 environmental regulation for new buildings ambitiously incorporates the challenge of summer comfort. Better-designed buildings will help limit the discomfort and danger associated with heat waves, as well as the use of air conditioning. This action could be complemented by the establishment of a common reference framework to go beyond the RE2020. Furthermore, a simplified summer comfort indicator has been integrated into the energy performance certificate.

Regarding transportation, the SNCF Réseau/SNCF Voyageurs and RATP heatwave plans have been strengthened and will continue to be so (prevention of incidents on railway infrastructure, prevention of embankment fires, prevention of the impact of heat on rolling stock, improvement of passenger information, provision of water in stations and on board trains, and improved working conditions for staff). In parallel, the passenger assistance agreements for exceptional situations, which have existed in several departments since the 2010s between SNCF Voyageurs and various organizations (prefectures, civil protection, French Red Cross, etc.), are being progressively taken over and updated by SNCF Réseau, which is responsible for coordinating crisis management on the railway network. These agreements, now signed by SNCF Réseau and approved civil security associations (Civil Protection, French Red Cross), allow SNCF Réseau to trigger, at its initiative and for the benefit of transported passengers, regardless of the passenger carrier (SNCF Voyageurs, Eurostar, Thalys, Trenitalia France...), additional resources to those of the carrier itself, subject to billing to the carrier concerned.

These resources can be activated year-round, including during heat waves. These agreements are standardized to provide the same contractual framework across all regions, while still allowing each region the freedom to choose its civil defense partners. An annual review of these agreements is also planned between SNCF Réseau and the railway companies that benefit from them.

Regarding the monitoring of water resources, a hydrological anticipation and monitoring committee has been set up in order to monitor and inform as early as possible about the state of the hydrological situation and the availability of water resources.

Beyond these measures already implemented, the following actions will be taken.

Action 1. Awareness-raising and communication campaign aimed at the general public

• Communication on minor repairs that can be carried out in housing

Heat waves are one-off climatic phenomena, a significant part of whose impact can be limited by simple actions (installation of blinds, shutters, planting vegetation, etc.).

Based on the Ademe guides "[Hot outside, cold inside – keeping your home cool in summer" and of the Anah](#) with regard to energy renovation operations in overseas (Guide to financial aid for renovation in overseas from Ademe, Guide France Rénov': <https://france-renov.gouv.fr/renovation/energetique-outre-mer>), information will be disseminated every year on possible works in housing, their cost, their level of difficulty (can be carried out by oneself or preferable recourse to a craftsman) and any existing financial aid, highlighting small works which can be carried out quickly.

Before periods of intense heat, the dissemination of this information will allow individuals to schedule these works in time and avoid the emergency use of very energy-intensive and inefficient portable air conditioners.

Pilot: Ministry of Ecological Transition and Territorial Cohesion

Partner operators: Ademe, Anah, France Rénov

- **Communication on appropriate behavior in one's home**

In homes, heat waves are occasional weather events, and their impact can be mitigated by simple actions (ventilation, closing shutters during the day, etc.). These best practices are not always known to the general public, particularly in regions unaccustomed to high temperatures, leading to the urgent purchase of energy-inefficient and ineffective portable air conditioners.

During heat waves, information campaigns will be conducted to remind people of the best practices to adopt in their homes. These campaigns may be adapted to the specific circumstances of the overseas territories, if necessary.

Pilot: Ministry of Health and Prevention

- **Information and accountability campaign for owners and holders animals**

In the event of high temperatures, animals may be exposed to conditions detrimental to their well-being and survival, due to insufficient awareness among their owners of the harmful effects of a heat wave and the existing means to limit these consequences, and due to a lack of awareness of their responsibilities specified in article L.214-1 of the rural and maritime fishing code.

A national information campaign will be conducted, targeting professionals in livestock and companion animal farming, animal transporters, and owners of wild animals (amateur or professional breeders, zoos, circuses, etc.). Finally, it will also target the general public regarding pets and equines, so that owners and keepers can take preventative and management measures to ensure the well-being of their animals during a heatwave.

This information campaign will be disseminated through the specialized press and/or through digital means relayed by professional livestock organizations, integrators, chambers of agriculture, livestock technical institutes, zoological park networks, representatives of pet shops, associations of breeders of non-domestic animals, the national commission of fairground and circus professions, local authorities, and state services for the benefit of the target population.

This information and responsibility campaign for owners and keepers of animals will consist of conveying appropriate messages throughout the year, outside the summer period, in the approach of summer, when climatic conditions are normal and in the pre-alert phase and during a heat wave.

Pilots: Ministry of Agriculture and Food Sovereignty, Ministry of Ecological Transition and Territorial Cohesion

- **Communication about the risk of drowning**

In France, accidental drownings are responsible for approximately 1,000 deaths each year, almost half of which occur during the summer months. The results of epidemiological surveillance ([drowning surveys](#)) (from [Public Health France](#)) show that high, even heatwave, temperatures lead to behaviors that very directly increase the risks (people swim more, they expose themselves more).

The ministries responsible for health and sports, in partnership with Public Health France, are carrying out awareness campaigns for several years on the risk of accidental drowning, which affects all ages and all swimming areas. These campaigns aim in particular to inform people about the correct behaviors to adopt to avoid drowning, as well as to promote a responsible approach to the aquatic environment by:

- the dissemination of [prevention tools](#) and a [communication kit](#), including, in particular, a poster on the [risks of drowning during periods of extreme heat](#): materials and messages broadcast in key locations for public information and spots on local radio stations;
- the creation of video tutorials to encourage an educational relationship between parents and children
children » around water accessible via the following
link <https://www.sports.gouv.fr/preventiondesnoyades>.

For the summer of 2023, the ministries in charge of health and sports, in partnership with the National Consumer Institute (INC), launched on May 17 and throughout the summer an unprecedented drowning prevention campaign focused on the right actions to adopt for effective supervision of swimming with children, under the slogan "[You care about them, don't take your eyes off them!](#)". This campaign includes a dedicated "[ConsoMag](#)" program, programmed on France Télévisions channels, as well as a series of 3 shocking testimonies from parents of children who have lost a child following a drowning.

Pilots: Ministry of Sports and the Olympic and Paralympic Games, Ministry of Health and Prevention

Partner organizations: Public Health France, National Consumer Institute

Action 2. Inventory of cool islands and public fountains in municipalities

In urban areas, temperatures are higher than in the surrounding countryside: this is the urban heat island effect. With climate change, urban heat islands will become more pronounced and risk creating very poor living conditions. The Green Fund

This allows us to finance actions for renaturation and combating urban heat islands.

Several major cities have begun to identify "cool islands" or "cool spots"—places where residents can take refuge for a few hours to cool down. These can be naturally cool places due to the presence of water or trees (parks, lakes, swimming pools), due to their thermal inertia (churches), or due to the presence of air conditioning or cooling systems (museums, cinemas, shopping centers). Public fountains offering free drinking water can also help residents better cope with periods of extreme heat.

Beyond making this information available locally by displaying it on the premises concerned, by signage or by digital information and any other suitable means of communication, the prefects will centralize information on the cool islands, cooled sites and public fountains available in the department, allowing people to quickly locate the cool space closest to them.

Reference: Guide "Coping with heat waves with your municipal emergency plan – Recommendations and best practices, Ministry of the Interior - Directorate General for Civil Security and Crisis Management, June 2021

Pilot: departmental prefects

Partner organizations: mayors

Action 3. Dissemination of prevention messages in transport

During periods of extreme heat, travellers entering an area under alert will be informed of health risks and the risk of forest and vegetation fires through the dissemination of recommended behaviour:

- in airports: posters, announcements in airports and on airplanes, including in foreign languages
- In major train stations: posters and announcements in stations by SNCF Gares & Connexions (reminding passengers to stay hydrated, refill their water bottles, providing information on water fountains or water available in retail outlets) and by railway companies (SNCF)
Passengers (Eurostar and Trenitalia) on trains (recommendations to stay hydrated, information on water availability at the bar), including in foreign languages
- on the RATP network (posters, multimodal information campaigns, etc.)
- on the national road network: use by managers (interdepartmental road directorates and motorway concession companies) of the tools at their disposal (messages, including in foreign languages, on variable message signs, radio messages on motorway FM, awareness campaigns on motorway rest areas, etc.)

Pilot: Ministry of Ecological Transition and Territorial Cohesion

Associated partners: SNCF Gares & Connexions, railway companies (SNCF Voyageurs, Eurostar, Trenitalia...), RATP, ADP, interdepartmental road authorities, motorway concession companies

Action 4. Initial assessment of the state-backed start-up "Fresher My City"

"Fresher My City" is a free public service offered by the French Agency for Ecological Transition (ADEME). Its mission is to help local authorities choose sustainable and long-lasting urban cooling solutions. It is a valuable decision-making tool available to local governments, but it is not widely known.

Pilot: Ministry of Ecological Transition and Territorial Cohesion

Partner operator: Ademe

Action 5. Dissemination of SMS messages by telephone operators in exceptional circumstances

FR-Alert is the new public alert and information system. Deployed nationwide since the end of June 2022, FR-Alert allows anyone with a mobile phone to be notified in real time of their presence in a danger zone, informing them of the actions to take to protect themselves.

FR-Alert will integrate the risks associated with heat waves for exceptional situations posing an imminent danger to the population, as a tool available to the director of operations (prefect) to prescribe specific protective measures tailored to each area. The Nord department tested such integration in April with an alert for a campsite.

Pilots: Ministry of the Interior and Overseas Territories, Ministry of Ecological Transition and Territorial Cohesion

Associate partner: Ministry of Health and Prevention

Action 6: RTE and Enedis strengthen their monitoring and intervention capabilities for periods of heat waves and drought

The bare conductors of overhead power lines expand due to ambient temperature and the current flowing through them, which can lead to a reduction in their height during periods of intense heat. To maintain safety distances between these power lines and the surrounding environment, electricity transmission may be reduced during heat waves to minimize this expansion. This reduction in transmission is not felt by consumers: consumption levels are sufficiently low in summer for RTE (the French electricity transmission network operator) to utilize network redundancies and distribute the load.

For some underground lines in the public network, heat waves can increase the risk of localized incidents that may, in some cases, necessitate the disconnection of the affected lines. Network operators are undertaking underground cable renewal programs to mitigate this risk by prioritizing the most exposed areas.

Furthermore, an accelerated adaptation strategy for overhead lines to rising temperatures is being studied as part of the next Ten-Year Network Development Plan (SDDR).

RTE and Enedis have a proven crisis management system in place to deal with events affecting overhead power lines (currently, however, the events most impacting the network are related to storms and snowfall, and to a lesser extent to heat waves), including emergency response resources deployed across the country. To address the increasing frequency and intensity of heat waves, Enedis will strengthen its monitoring and response capabilities for periods of heat waves and drought starting in the summer of 2023.

Additional actions:

• Communication aimed at foreign tourists

Non-French-speaking populations, including tourists, are identified as being less aware of the health risks associated with high temperatures, probably because the usual communication materials are not being conveyed through the appropriate communication channels.

Embassies will be informed of the arrival of a heat wave so that they can inform their citizens.

Pilot: Ministry for Europe and Foreign Affairs

Associated partners: Ministry of Health and Prevention, Ministry of the Interior and Overseas Territories
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Axis 2. For younger people and vulnerable individuals

Before and during heat waves, actions can be taken to protect young people and vulnerable people.

Regarding young people, several best practices for reducing the impact and consequences of heat waves have already been implemented. To minimize thermal discomfort in school buildings during hot periods, the Ministry of National Education and Youth has published a national guide listing and describing actions to mitigate the impact of heat waves. These actions require only limited investment and can be carried out before or during heat waves. The guide takes into account the diversity of educational institutions. Furthermore, to harmonize decision-making during heat waves, two fact sheets have been developed to assist in deciding whether to close primary schools or other group activities for minors.

Regarding vulnerable individuals, since 2004, mayors have been required to establish a register of elderly and disabled residents of their municipality living at home who request to be included. The purpose of this register is to enable targeted intervention by health and social services in the event of an alert or emergency plan being activated. Few vulnerable individuals have registered on these registers. Therefore, in the lead-up to the heatwave season, it is crucial to improve the registration rate on the municipal registers.

Action 7. Mobilizing young people in universal national service to register vulnerable people on municipal registers and in support of climate change awareness actions

Under Law No. 2004-626 of June 30, 2004, concerning solidarity for the autonomy of elderly and disabled persons, mayors are required to establish a register of elderly and disabled persons living at home in their municipality who request to be included. The purpose of this register is to enable targeted intervention by health and social services in the event of an alert and emergency plan being activated. Three categories of persons may be included on this register: persons aged 65 and over, persons over 60 recognized as unfit for work, and persons recognized as disabled. During heat waves, municipal services regularly call registered individuals to check on their well-being.

The difficulty lies in the availability of staff during holiday periods.

According to a 2015 survey, while preventative measures seem to be well known, awareness of one's own risks is low, even among people over 65 (only 4% of the elderly believe they are at significant risk during a heatwave and have registered with their local authorities). It is therefore essential to improve the identification of vulnerable individuals in these local registries.

To this end, young people participating in the National Universal Service (SNU) can be mobilized to encourage vulnerable individuals to register. The SNU is open to young people of French nationality aged 15 to 17. It includes a two-week cohesion phase carried out in another region and a general interest mission of at least 84 hours, completed (either continuously or intermittently) near their home. Each young person can then continue with a minimum three-month commitment period in the department of their choice.

Throughout the year, young people participating in the SNU (National Universal Service) will be able to carry out their community service mission in town halls and conduct public awareness campaigns on the causes of climate change and its impacts. This can be done by using electoral lists or lists of home meal delivery services.

For example, they will also be able to call people over 65 and encourage them to register on the municipal registers, check on them and give them advice.

For young people participating in the National Universal Service (SNU), a commitment contract will be signed between the town hall, the State (represented by the departmental youth, engagement, and sports service), and the young people's legal guardians. They will support staff, not replace them, and cannot be placed in a position of direct responsibility. As such, a town hall staff member will accompany the young people during their mission as a mentor. (Explanations and the necessary procedures are detailed on the webpage: <https://www.snu.gouv.fr/mode-emploi-pour-les-responsables-de-structure-comment-proposer-des-missions-dinteret-general/>).

Individuals undertaking civic service missions within the framework of the Senior Solidarity Civic Service (SC2S), whose mission is to combat social isolation and who receive specific training on outreach and working with older adults developed by SC2S stakeholders, may be mobilized in this context. When they carry out this mission within an organization involved in the heatwave plan, provided that this activity is specified in their commitment contract, they may participate alongside staff mobilized for this purpose in actions aimed at encouraging and supporting vulnerable individuals to register.

communal.

Pilots: Ministry of National Education and Youth, Civic Service Agency

Associated partners: Ministry of Health and Prevention, Ministry of Solidarity, Autonomy and Disabled Persons

Action 8. Raising awareness among guardians/curators of the need to register vulnerable persons on municipal registers

Guardianship and curatorship are two legal measures designed to protect a person and their assets. The procedures for applying for guardianship or curatorship are very similar, but their effects differ: a person under curatorship remains autonomous in simple daily tasks but must be assisted with important decisions; guardianship, on the other hand, is a more protective measure for individuals who have lost their autonomy and need representation in legal matters. When no family member or close relative can assume guardianship or curatorship, the judge appoints a court-appointed guardian for adults, selected from a list compiled and maintained by the prefect of the department.

Every year, an awareness campaign will be conducted for guardianship judges and court-appointed representatives to encourage curators and guardians to pay particular attention to individuals under protective measures who are isolated and to ensure their registration in municipal records. Supporting information will be provided by the Ministry of Solidarity, Autonomy and Persons with Disabilities.

Pilot: Ministry of Justice

Associated partner: Ministry of Solidarity, Autonomy and Disabled Persons

Action 9. Implementation of an annual inspection of cooled rooms/premises as well as summer comfort equipment in schools and nurseries and examination rooms

To minimize thermal discomfort in school buildings during hot periods, a national guide listing and describing actions to limit the impact of heat waves, requiring only limited investment and which can be carried out before or during heat waves,

Guidelines, taking into account the diverse contexts specific to each school and institution, have been disseminated by the Ministry of National Education and Youth. These actions include both:

- behaviors to adopt
- maintenance work or actions to be carried out
- the creation of cool spaces.

This guide is available here: <https://batiscolaire.education.gouv.fr/guide-vagues-de-chaleur-240301>.

The recommendations of the guide will be reiterated every year.

The check of the proper functioning of the equipment ensuring summer comfort (window openings, presence/ functioning of blinds, VMC, presence of a cooled room in schools, etc.) will be carried out every year in the spring in all schools and nurseries, under the responsibility of the head of the establishment or the director of the establishment in conjunction with the owner of the buildings (municipalities in particular).

Pilot: mayors, local authorities

Associated partners: Ministry of National Education and Youth, Ministry of Agriculture and Food Sovereignty (School Education), Ministry of Solidarity, Autonomy and Disabled Persons, Ministry of Ecological Transition and Territorial Cohesion

Action 10: Implementation of a control system for rooms and refreshed premises that can accommodate examinations

To minimize exam postponements during hot periods due to thermal discomfort in school buildings, an inspection of air-conditioned rooms and spaces near middle and high schools that are large enough to accommodate the Brevet and Baccalaureate exams will be carried out annually in each academic region.

Where necessary, funds may be allocated for their rental.

Pilot: Ministry of National Education and Youth

Partner organizations: Ministry of Agriculture and Food Sovereignty (school education), municipalities, inter-municipal public cooperation establishments with their own tax base

To go further

Beyond these measures, various actions will be taken for younger audiences, in order to move towards the systematic presence of a cooled room in schools, or to help with the decision regarding the postponement of exams.

Axis 3. For workers and businesses

Heat waves can impact businesses, their operations, and workers. Actions can be taken before and during heat waves to mitigate and adapt to these impacts, whether through workplace adjustments or renovations, or by adopting best practices.

A fact sheet on energy saving certificates has already been published to encourage work improving summer comfort (reflective paint on the roofs of buildings used for commercial purposes).

Action 11. Creation of a guide for businesses regarding work that can be carried out in offices and communication on appropriate office behavior.

• Creation of a guide for businesses regarding the work that can be carried out in the offices

In the short term, heat waves are one-off climatic phenomena, a significant part of whose impact can be limited by simple actions (installation of blinds, planting vegetation, etc.).

Similar to the guide "[Hot outside, cold inside – keeping your home cool in summer](#)"

Published by ADEME in 2018, information on simple, quick improvements to office temperature will be disseminated. This information will include details on the cost and duration of these projects, as well as any available financial assistance. This communication, coming before the heatwave, will allow businesses to schedule these improvements in advance.

The National Institute for Research and Safety for the Prevention of Occupational Accidents and Diseases (INRS) has published several resources offering practical advice on the design and layout of workplaces and workstations to combat heat-related risks:

- ED950 "[Design of workplaces and work situations](#)" » provides recommendations on the thermal environments;
- ED6371 provides advice on the layout of workstations, such as technical adaptations (fan, misting system, blind, outdoor shelter...), the use of air conditioning, the provision of air-conditioned rest areas...
- ED 6096 is intended for occasional project owners: business leaders, local elected officials, and other contracting authorities. Its objective is to help them integrate the fundamentals of occupational health and safety into their projects, including those related to air quality and thermal comfort.

Pilot: Ministry of Ecological Transition and Territorial Cohesion

Associated partner: Ministry of Labour, Full Employment and Integration

• Communication on appropriate office behavior

During periods of heat waves, workers must adapt their behavior to prevent heat-related risks. Throughout the seasonal monitoring period, the Ministry of Labour, Full Employment and Integration, the regional directorates for the economy, employment, labour and solidarity, and prevention organizations communicate with workers and make available

Educational materials outlining the measures to be implemented are available. Agents from the Labour Inspectorate system are also mobilized to advise and raise awareness among workers.

The occupational health and safety inspectorate and occupational health and safety prevention services (SPST) may also be mobilized.

Regarding sedentary work: in the short term, in offices, heat waves are occasional climatic phenomena whose impact can be partially limited by simple actions.

(ventilation, closing blinds during the day, etc.). These good practices are not always known to the general public, especially in regions not used to high temperatures.

During heat waves, information campaigns will be carried out to remind people of the correct practices to adopt in their offices.

Pilot: Ministry of Labour, Full Employment and Integration

Associate partner: Ministry of Health and Prevention

Action 12. Strengthening of labour inspection controls

During a heat wave, labor inspection controls will be stepped up.

The mobilization of services is the subject of an instruction distributed to the regional directorates of the economy, employment, labor and solidarity (DREETS) each year, before the start of the vigilance period.

Throughout the alert period, inspectors are instructed to conduct targeted inspections of businesses in the sectors most affected. Particular attention will be paid to activities that expose workers to greater heat-related risks, such as outdoor work (construction, agricultural work), but also restaurants, bakeries, dry cleaners, etc. If the alert level is raised to red, these inspections will be intensified.

The inspections will serve as a reminder of the obligations regarding risk assessment and the need to adapt preventive measures to the circumstances and improve existing practices, through updating the single risk assessment document (articles L. 4121-1 et seq., and R. 4121-1 of the French Labor Code). Employers will be asked to adapt work organization in anticipation of extreme heat. All employer obligations are outlined in fact [sheet O2/K](#). from the ORSEC guide –
specific provision for "health management of heat waves".

Depending on their findings, the inspectors may invoke the provisions relating to the layout of workplace premises, which include:

- the obligation to renew the air in the workplace (articles R. 4222-1 and following of the labor code),
- the obligation to provide drinks (article R. 4225-2 et seq.),
- the obligation to protect workers from climatic conditions, as far as possible, when they are employed outdoors (Article R. 4225-1 of the Labour Code).

These provisions may give rise to the notification of a request for verification and/or a formal notice prior to the official report.

In the event of a dangerous situation resulting from non-compliance with the general principles of prevention (articles L. 4121-1 to L. 4121-5, L. 4522-1 and L. 42221-1 of the labour code), the control agents may transmit to the DREETS a report with a view to the notification of a formal notice (article L. 4721-1).

Particular attention will be paid to the employment of young workers. Indeed, employers are prohibited from assigning young people to work involving exposure to extreme temperatures likely to harm their health (Article R. 4153-36 of the French Labour Code); there are no exceptions to this prohibition.

If an inspector observes the employment of a young person in such a situation, they may issue a notice of withdrawal of the young person assigned to prohibited work (Article L. 4733-2 of the French Labor Code). Failure to comply with this decision is punishable by an administrative fine (Article L. 4752-1 of the French Labor Code). Furthermore, employing a young person in prohibited work is punishable by a fine (Article L. 4753-2 of the labor code).

Pilot: Ministry of Labour, Full Employment and Integration

Action 13. Restriction of the transport of live animals and management of peak activity by rendering companies

During heat waves, the loading, transport, and unloading of livestock can cause respiratory distress, dehydration, and stress, endangering their lives. The ministerial decree of July 22, 2019, restricting the road transport of live terrestrial vertebrates during heat waves, imposes restrictions on the transport of live terrestrial vertebrates during these periods. It prohibits the road transport, for commercial purposes, of live terrestrial vertebrates between 1 p.m. and 6 p.m. during heat waves published on the Météo-France website (<https://vigilance.meteofrance.com/>). Transport schedules must be based on the latest heat wave map published the day before transport. Exemptions are possible when the vehicle is equipped with air conditioning or a dual ventilation and misting system to regulate animal temperatures, if the transport involves three animals or fewer, or when transport is necessary for veterinary or animal welfare reasons. Furthermore, heat waves can lead to excessive livestock mortality and deterioration of collected carcasses, potentially causing delays or even a failure to collect carcasses from farmers and overstocking of carcasses under unsuitable conditions on farms and/or at rendering plants.

When a heat wave is announced, rendering companies will be put on alert in order to anticipate a possible peak in activity and to be able to adapt their organization accordingly.

Pilot: Ministry of Agriculture and Food Sovereignty

Additional actions

• Organization by Voies Navigables de France (VNF) of a meeting of local commissions users

The decrease in river flows and the failure to fill the reservoir dams of Voies Navigables de France (VNF), due to a lack of rainfall, may lead VNF to take specific measures to restrict and stop river navigation and water distribution.

The systematic organization by VNF, prior to periods of risk, of a meeting of local user committees to review the hydrological situation and the risks of closures

canals will allow us to anticipate problematic situations and find solutions (delayed opening of canals, anchoring restrictions, alternative routes, etc.).

VNF will involve inland port managers as needed, depending on the possible impact of these risky episodes on port operations.

Pilot: VNF

- **Informing employers of the upcoming heat wave**

Working in hot conditions, especially outdoors, poses health risks to employees and leads to workplace accidents. Employers are obligated to take the necessary measures to ensure the safety and protect the health of their workers.

The announcement of an upcoming heat wave is relayed to employers, at national and regional levels, so that they can put in place the appropriate measures to protect their employees.

Throughout the seasonal monitoring period, the Ministry of Labour, Full Employment and Integration, the Regional Directorates of Economy, Employment, Labour and Solidarity, and the prevention bodies communicate with employers and make available educational documents reminding them of the measures to be implemented.

Labor inspectors are also mobilized to monitor and support businesses in addressing heat risks. They are specifically targeting sectors particularly exposed to heat risks (agriculture, construction, hospitality, etc.). Occupational health inspectors and occupational health and safety services are also involved.

Pilot : Ministry of Labour, Full Employment and Integration

- **Regulation of certain activities according to the level of fire risk by the mayor and the prefect**

As a reminder, the mayor, within the territory of his municipality, and the prefect of the department, for all or part of the departmental territory (over several municipalities), take, respectively, within the framework of their general administrative police powers, any regulatory measure adapted and proportionate to the circumstances.

These measures aim to prevent the risk of fire during periods of intense heat and associated drought. They can also be mitigated by taking into account other specific weather conditions, such as periods of wind.

These measures may, in particular, aim to:

- restrict access to sensitive natural areas for people and vehicles;
- restrict or limit in time and space professional and leisure activities that may accidentally generate fires, for example: temporary ban on mechanical mowing, temporary ban on the use of solid fuel barbecues, temporary ban on smoking near natural areas...;
- supplement, where necessary, the permanent municipal regulations relating to domestic burning.

Pilots: prefect of the department, mayor

To go further

Various measures will be put in place in the longer term to strengthen measures to limit the impact of heat waves on the conditions for carrying out economic activity and for workers:

During heat waves, it is advisable to stagger working hours, particularly for construction workers, to avoid exposing them to the hottest hours. Conversely, to minimize noise exposure for local residents, municipal decrees prohibit construction sites from operating early in the morning. A guide will be developed to help mayors adapt these decrees to allow for adjusted work schedules and earlier start times during periods of extreme heat.

- In order to ensure worker safety while maintaining maximum continuity of

Given the economic activity in France, certain public and private companies may be required to develop a heatwave plan. For companies already equipped with a business continuity plan, such as operators of vital importance, this would involve amending it, where necessary, to incorporate the risk of heatwaves.

Axis 4. For athletes and spectators of cultural events

Heat waves can have a significant impact on sporting and cultural events organized during the summer months: postponements, cancellations, etc. To best anticipate the organization of these events and take into account the heat risk, various prevention and information measures can be implemented for organizers and participants.

Action 14. Departmental census of sporting and cultural events

From June to September, local federations, municipalities, and associations organize sporting and cultural events that attract large crowds to outdoor gatherings, sometimes during the hottest hours of the day. Aside from major recurring events such as music and theater festivals or the Tour de France, departmental prefects are not always aware of these events, which can create management problems, particularly when it comes to informing organizers about the potential consequences of an impending heatwave. Furthermore, these events are often organized by less experienced volunteers.

In the spring, a census of these events will be carried out in each department. Mayors will be asked to send the prefect a list of events planned in their territory and the contact details of the organizers.

Pilots: departmental prefects

Partner organizations: mayors

Action 15. Invite organizers of sporting and cultural events to raise their awareness of best practices in the event of heat waves

The organizers of these events will then be brought together by the prefects to raise awareness of the risks associated with high temperatures, remind them of the instructions and procedures to follow and anticipate the measures that would need to be put in place in the event of a heat wave.

When a heat wave is announced, the organizers of sporting or cultural events will be notified by the prefects of the department to adapt their organization and put in place the measures that will have been discussed beforehand.

Pilots: departmental prefects

Partner organizations: mayors

Additional actions:

• Information for sports educators

During previous heat waves, while people over 75 were the most affected, the 15-44 age group was also significantly impacted. During some episodes, young people and those of working age were more affected, likely due to the continuation of risky activities and behaviors, particularly sports during the hottest hours, despite safety guidelines.

Each year, at the beginning of the seasonal monitoring period, the sports directorate sends general and sports-specific health recommendations by email to regional offices, national public institutions under its supervision (INSEP, CREPS, schools), and sports federations.

In case of extreme heat. When a heatwave is announced, an information email will be sent to all 156,000 sports educators.

Pilot: Ministry of Sports and the Olympic and Paralympic Games

Associate partner: Ministry of Health and Prevention

Focus - Preparing stakeholders in the electricity sector

Even though summer consumption levels are much lower than winter levels, and therefore the risks to the electrical grid are less in summer than in winter, heat waves generally result in increased energy consumption compared to the rest of the summer, particularly for air conditioning. At the same time, the availability of some generation capacity can be reduced, notably because many generation facilities are undergoing maintenance.

However, it is important that messages in favor of energy sobriety are disseminated during these heat wave episodes, starting with setting a reference temperature for air conditioning above 26°C, or, when going on holiday, unplugging all electrical appliances that can be unplugged.

The network's operating methods are the same as the rest of the year.

• Availability of the nuclear reactor fleet

The increase in the temperature of rivers upstream of power plants, or the decrease in the river flow, can lead to limiting or even stopping their production when the water discharged exceeds the temperature threshold and the flow downstream is too low.

These production adjustments are not related to nuclear safety but to environmental protection, and the effects on electricity production are very marginal on average over the year.

The recurrence of periods of rising average or extreme temperatures has led for several years to the implementation of special monitoring in summer and EDF, operator of the power plants, is responsible for taking the necessary measures to take into account the consequences of climate change, under the control of the nuclear safety authority (ASN).

In the event of strain on the electrical grid, and at the request of the transmission system operator, EDF may, under certain conditions, deviate from discharge temperatures to ensure a balance between supply and demand. Operationally, the organizational structure adopted to address these situations allows for a rapid and proportionate response.

- Within the framework of applicable regulations, EDF may occasionally resort to increased levels of thermal discharges in the event of high temperatures, under the control of the ASN.
- In exceptional circumstances where this proves insufficient, if the State identifies a public need to continue the operation of certain reactors, the ASN may grant an emergency exemption from the applicable requirements.

• Hydroelectricity

Heat waves can directly affect hydroelectric facilities. Indeed, heat wave episodes can lead to an increase in glacial meltwater input, requiring maintenance work on hydroelectric facilities and more frequent monitoring of the installation's condition.

However, the effect is also indirect for the structures, their operators and the electrical system:

- Overcrowding of waterways can also be observed, requiring active communication from dam operators to the various users about the risk of water level variation.
- For operators, heat wave episodes generally lead to increased water consumption to support low water levels and downstream uses (particularly due to increased evapotranspiration), even though electricity needs are more limited than in winter and these periods normally allow the resource to be preserved for electricity production, except when RTE calls on these production facilities to ensure the balance between supply and demand.

- **Annual preparation**

To enable effective decision-making, it is necessary to cross-reference relevant information on energy supply and electricity grid management, on climatic conditions and the projected duration of heat waves, and on the impact of heat emissions on the environment and the control programs to be implemented. This information is available from various stakeholders who must be contacted quickly, sometimes within a few days. Furthermore, lessons learned from previous years or the specific circumstances of a given season (for example, in terms of the availability of different electricity generation resources or the situation in neighboring European countries) may lead to adjustments in operating procedures for the coming season.

The Ministries of Energy Transition and Ecological Transition and Territorial Cohesion organize every year, before the summer, a meeting with stakeholders in the electricity sector and the relevant public establishments (Météo France, OFB, etc.) in order to review seasonal forecasts and share procedures in case of a heat wave and the methods of disseminating the various information bulletins.

Pilots: Ministry of Energy Transition, Ministry of Ecological Transition and Territorial Cohesion